lab08

```
library(reticulate)
py_require("polars")
```

```
import polars as pl

kw = dict(separator="\t", null_values="\\N", quote_char=None)

basics = pl.read_csv("title.basics0.tsv.gz",
                     columns=["tconst", "titleType", "primaryTitle", "genres"], **kw)
ratings = pl.read_csv("title.ratings.tsv.gz", **kw)
principals = pl.read_csv("title.principals0.tsv.gz",
                         columns=["tconst", "nconst", "category"], **kw)

filmes = (
    basics
    .filter(pl.col("titleType") == "movie")
    .drop("titleType")
    .join(ratings, on="tconst", how="inner")
)
```

1.

```
top5 = (
  filmes
  .sort(["averageRating", "numVotes"], descending=[True, True])
  .head(5)
)
print(top5)
```

```
shape: (5, 5)
```

tconst	primaryTitle	genres	averageRating	numVotes
str	str	str	f64	i64
tt33075815	Kaveri	Action	10.0	1023
tt27815015	Kurukku	Crime,Mystery,Thriller	10.0	451
tt28450376	Jedal Dar Omghe 30 Metri	Documentary	10.0	142
tt27859713	Sargashte	null	10.0	134
tt33507675	Gorgeous Rascal	Thriller	10.0	115

2.

```
genero_top = (
  filmes
  .filter(pl.col("averageRating") > 8)
  .with_columns(pl.col("genres").str.split(","))
  .explode("genres")
  .drop_nulls("genres")
  .group_by("genres")
  .agg(pl.len().alias("n_filmes"))
  .sort("n_filmes", descending=True)
  .head(1)
)
print(genero_top)
```

```
shape: (1, 2)
```

genres	n_filmes
str	u32
Documentary	10564

3.

```
bem_avaliados = filmes.filter(pl.col("averageRating") > 7.5).select("tconst")

top_atores = (
  principals
  .filter(pl.col("category").is_in(["actor", "actress"]))
  .join(bem_avaliados, on="tconst", how="semi")
  .unique(["tconst", "nconst"])
  .group_by("nconst")
  .agg(pl.len().alias("n_filmes"))
  .sort(["n_filmes", "nconst"], descending=[True, False])
  .head(3)
)
print(top_atores)
```

```
shape: (3, 2)

 nconst      n_filmes
 ---         ---
 str         u32

 nm0004660   199
 nm0595934   155
 nm3183374   124
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-10-14 10:44:32 -03"
```